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Lecture 4

—

Advanced Regression

Kaisermantel (*Argynnis paphia*)

- ◆ Der Kaisermantel oder Silberstrich ist ein Schmetterling (Tagfalter) der Gattung *Argynnis* aus der Familie der Edelfalter (Nymphalidae). Das Artepitheton leitet sich von *Paphia*, einem Beinamen der Aphrodite aus der griechischen Mythologie ab.
- ◆ Er wurde zum Schmetterling des Jahres 2022 gekürt.
- ◆ Die Falter erreichen eine Flügelspannweite von 55 bis 65 Millimeter in Mitteleuropa. Unterseitig sind die Flügel von Kaisermänteln grünlichweiß mit einem Hauch von Silber.
- ◆ Der Kaisermantel bewohnt naturnahe Wälder, Waldränder und neuerdings auch des Öfteren Gärten und Gebüsche. Die Männchen sind oberseitig intensiv orange gefärbt mit einem schwarzbraunen Muster auf den Flügeln. Weibchen sind etwas bräunlicher.
- ◆ Die Tiere ernähren sich vom Nektar von Disteln, Dost und Brombeeren, aber auch Sommerflieder.

Quellen:

* <https://de.wikipedia.org/wiki/Kaisermantel>

* <https://www.plantura.garden/gruenes-leben/schmetterlinge-die-10-beliebtesten-heimischen-schmetterlingsarten>



Advanced Regression

1. Polynomial Regression
2. Regularization – Ridge Regression
3. Regularization – Lasso Regression
4. Regularization – Elastic Net Regression
5. Further Regression Algorithms



Recap: Multiple Linear Regression

- ♦ **Multiple**: the scalar variable y is dependent on multiple values $x_{i1}, x_{i2}, \dots, x_{ik}$:

$$y_i = \beta_0 + x_{i1}\beta_1 + x_{i2}\beta_2 + \dots + x_{ik}\beta_k + \varepsilon_i$$

leading (for n data points) to the following equations:

$$y_1 = \beta_0 + x_{11}\beta_1 + x_{12}\beta_2 + \dots + x_{1k}\beta_k + \varepsilon_1$$

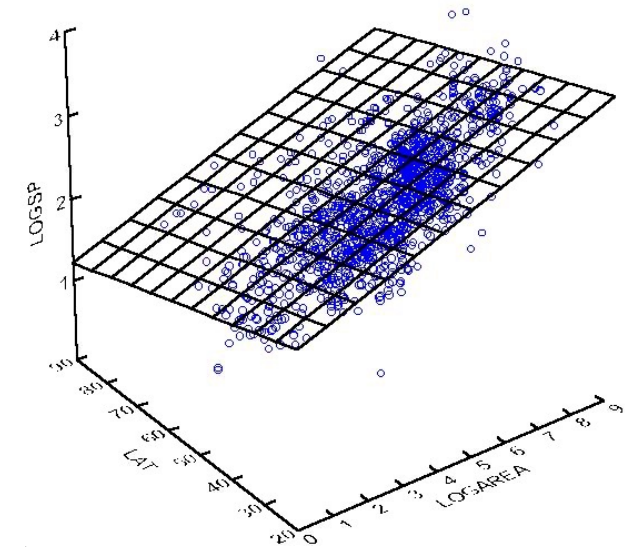
$$y_2 = \beta_0 + x_{21}\beta_1 + x_{22}\beta_2 + \dots + x_{2k}\beta_k + \varepsilon_2$$

...

$$y_n = \beta_0 + x_{n1}\beta_1 + x_{n2}\beta_2 + \dots + x_{nk}\beta_k + \varepsilon_n$$

which can be written more concisely as $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with

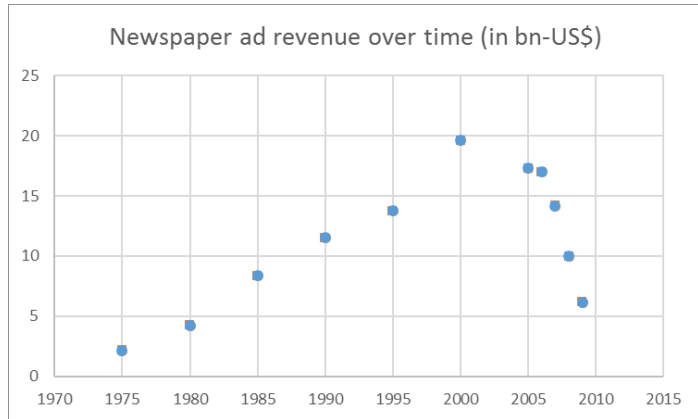
$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix} \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix} \quad \text{and} \quad \mathbf{X} = \begin{pmatrix} 1 & x_{11} & x_{12} & \dots & x_{1k} \\ 1 & x_{21} & x_{22} & \dots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{nk} \end{pmatrix}$$



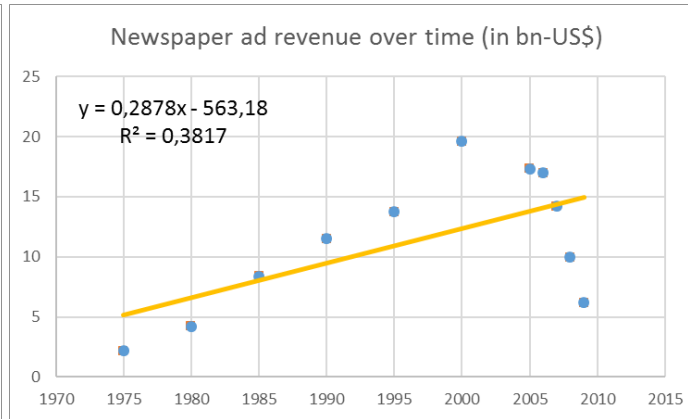
source: <http://ordination.okstate.edu/MULTIPLE.htm>



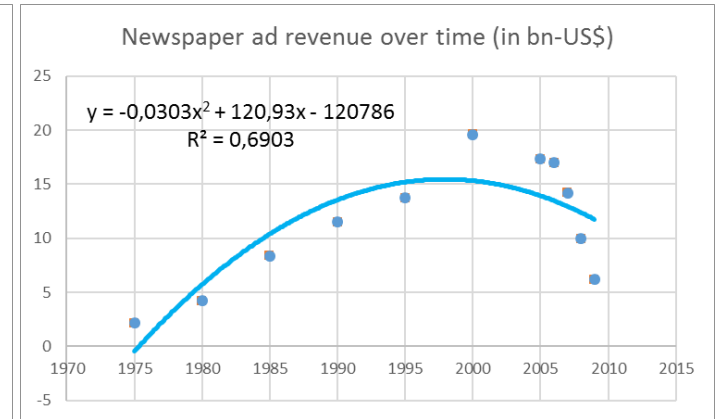
Not every dataset can be fitted nicely with a linear regressor



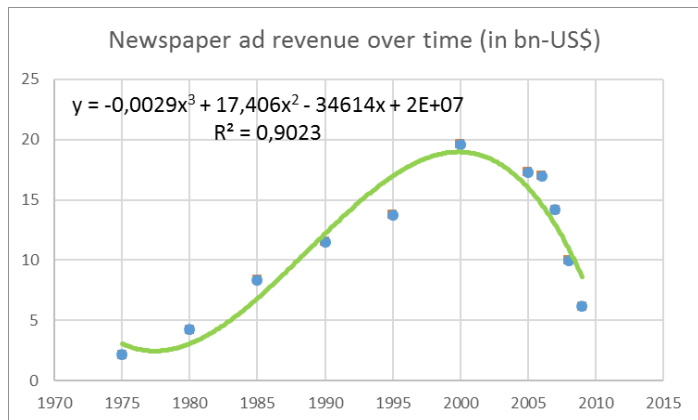
original dataset



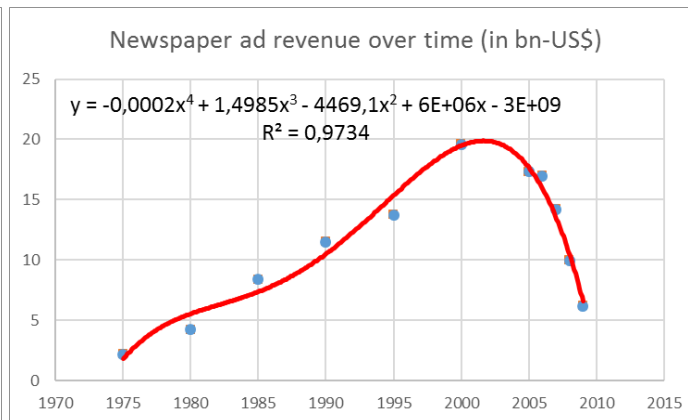
linear regression (x)



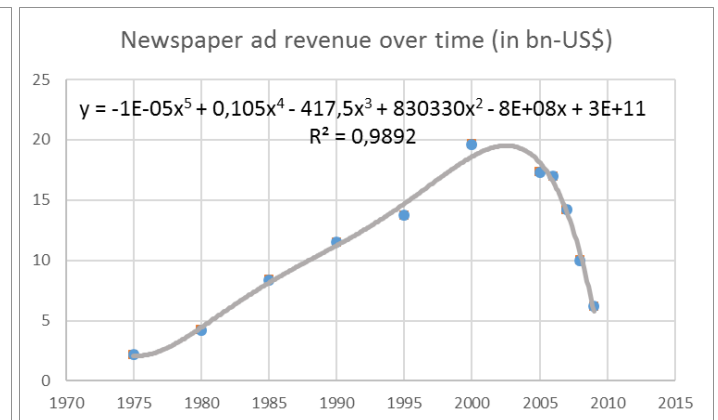
quadratic regression (x, x²)



polynomial regression of degree 3



polynomial regression of degree 4



polynomial regression of degree 5



Polynomial regression with one independent variable

- Instead of the the function $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ used in the Simple Linear Regression, use a **polynomial function of degree d** :

$$\begin{aligned} \text{degree 2 : } y_i &= \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \varepsilon_i & (\text{quadratic model}) \\ \text{degree 3 : } y_i &= \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3 + \varepsilon_i \\ \dots & \\ \text{degree } d : y_i &= \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3 + \dots + \beta_d x_i^d + \varepsilon_i \end{aligned} \quad \times \quad \begin{matrix} x_i & x_i^2 & x_i^3 \\ 1 & 2 & 4 & 8 \\ 1 & 5 & 25 & 125 \\ 1 & 3 & 9 & 27 \\ 1 & 10 & 100 & 1000 \end{matrix}$$

- This can be transformed into multiple linear regression: simply treat x , x^2 , ... as distinct variables, e.g. for degree 3, replace x_i by x_{i1} , x_i^2 by x_{i2} and x_i^3 by x_{i3} , leading to

$$y_i = \beta_0 + x_{i1}\beta_1 + x_{i2}\beta_2 + x_{i3}\beta_3 + \varepsilon_i$$



Polynomial regression with multiple independent variables

- Polynomial regression can easily be extended to multiple independent variables, by using all combinations of features and polynomial terms.

- Example:** y is dependent on two features x_1 and x_2 . Fit a (standard) polynomial of degree 2:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i1}^2 + \beta_4 x_{i2}^2 + \beta_5 x_{i1}x_{i2} + \varepsilon_i$$

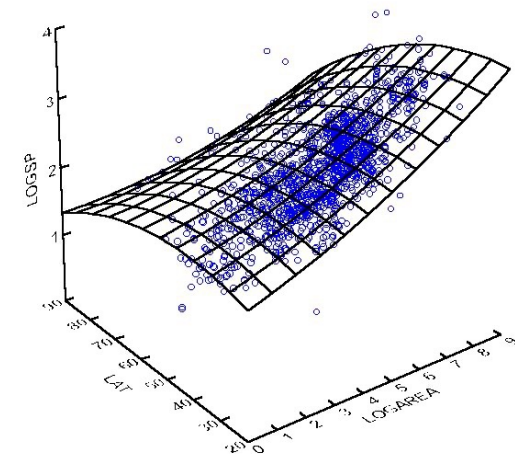
sum of all possible combinations (see example below)

→ multiple linear regression in 6 parameters.

- Beware:** With n features and polynomial terms up to degree d , the number of parameters is $\binom{n+d}{d} = \frac{(n+d)!}{n!d!}$. This can explode even when both n and d are moderate, e.g.,

- $n = 10, d = 3 \Rightarrow 286$ parameters
- $n = 10, d = 6 \Rightarrow 8008$ parameters
- $n = 50, d = 3 \Rightarrow 23426$ parameters

- Solution/Optimization identical to multiple regression (OLS)!**



source: <http://ordination.okstate.edu/MULTIPLE.htm>

e (c, m, a) = 0 +	1c + 2m + 3a +	4c^2 + 5ca + 6cm + 7m^2 + 8ma + 9a^2	10c^3 + 11c	length 0
				length 1
				length 2
				length 3



Exercises

Exercise 1

Cats and Mice





Advanced Regression

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Over-Fitting and Regularization

- ♦ Regression has **high-variance** if there are many weights β , which is often the case due to
 - Many features → high probability of irrelevant or correlated features
 - High-degree polynomial regression

→ Overfitting

- Reduce model complexity via **Regularization** necessary
 - **Ridge** Regression
 - **Lasso** Regression
 - **Elastic-Net** Regression



Ridge (L2) Regression

- ◆ Add a **penalty** equivalent to **square of the coefficients** to the cost function:

$$\mathbf{b} = \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \left(\frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2 \right) = \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \left(\frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \sum_{i=1}^k \beta_i^2 \right)$$

equiv.: f(x)
but vector here

upper 2: squared
lower 2: euclidean norm root (2)

where λ is a positive constant.

is a hyperparameter that needs to be tuned before training

penalizes big the bigger constant, the more penalized will choose

- ◆ Solution still possible with OLS leading to the estimator

$$\mathbf{b}_\lambda = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$$

theoretically, multiple lambdas would lead to more control (can specifically target certain features)
but is too complicated/expensive so this simplified version as presented here is used instead



Properties of Ridge Regression

- ◆ Penalty term λ regularizes the coefficients
 - Large values for coefficients are penalized.
 - When $\lambda \rightarrow 0$, cost function becomes the linear regression cost function.

- ➔ Ridge regression shrinks the coefficients
 - Reduces model complexity and multi-collinearity

- ◆ Multiple Regression is scale invariant, Ridge regression is NOT!

➔ Always normalize the variables so that they approx. have the same scale

➔ Note: coefficients may become small, but usually not 0

scale invariant:

can change scale of features which will reflect back to the beta constants, e.g. length in m is changed to cm

beta is scaled by inverse factor (i.e. feature 10 -> 100 would reduce (divide) beta by that factor)

ridge regression depends on the betas, so it is not scale invariant
a beta factored by 100, has 100 more effect than previously



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Lasso (L1) Regression

- ♦ **Lasso** (least absolute shrinkage and selection operator): very similar to Ridge Regression
- ♦ Add a **penalty** equivalent to **absolute value of the coefficients** to the cost function:

$$\mathbf{b} = \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \left(\frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1 \right) = \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \left(\frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \sum_{i=1}^k |\beta_i| \right)$$

einsernorm (betrug) anstatt quadrat

small changes in beta are not penalized as strongly

where λ is a positive constant.

- ♦ No closed form solution possible (cost function not differentiable), but all convex-optimization algorithms (e.g., SGD) can be used



Properties of Lasso Regression

- ◆ Penalty term λ regularizes the coefficients
 - Large values for coefficients are penalized.
 - When $\lambda \rightarrow 0$, cost function becomes the linear regression cost function.
- ➔ Lasso regression shrinks the coefficients and can shrink all the way to 0
 - Reduces model complexity
 - Selects relevant features
- ◆ Multiple Regression is scale invariant, Lasso regression is NOT!
 - ➔ Always normalize the variables



Ridge vs. Lasso Regression

- ◆ Both Lasso and Ridge can reduce overfitting
 - ability to reduce weights to 0
- ◆ Lasso can lead to feature selection, while Ridge shrinks only close to 0
- ◆ Ridge Regression works great
 - For small datasets: reduces sensitivity to the training data
 - If most features are usefull
- ◆ Lasso works great
 - If there are many useless features
- ◆ Lasso leads to simpler, easier to understand models

Dimension Reduction of Feature Space with LASSO

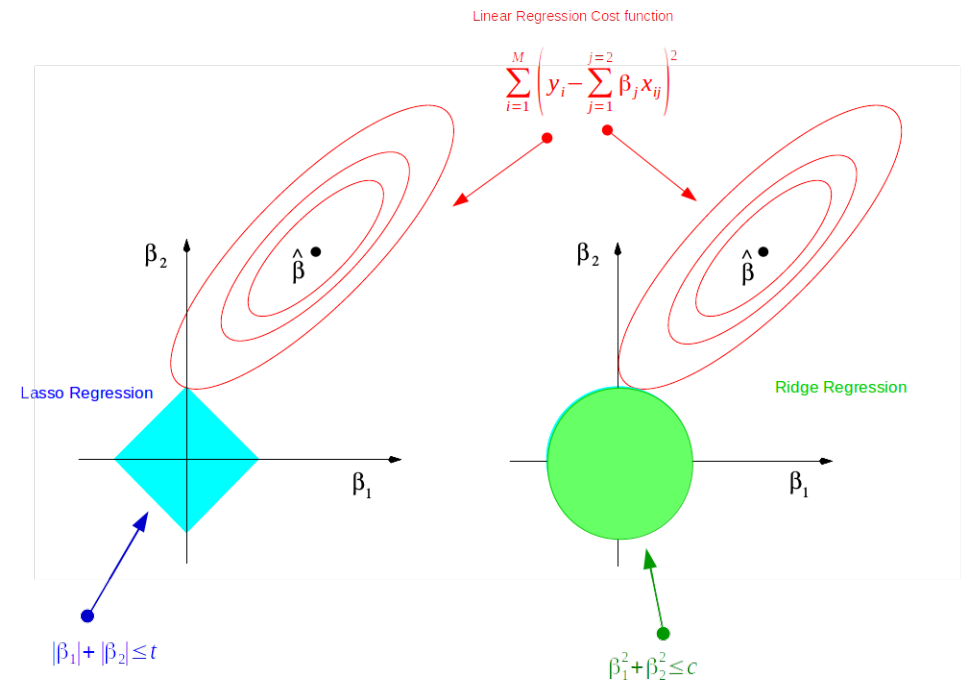


Fig: Why Lasso can reduce weights to 0 and Ridge usually cannot.
 schnittpunkt zwischen regression (kreis/diamant) und ellipse wäre idealwert

ridge schnittpunkt mit ellipse an versch. orten möglich, lasso schnittpunkt meistens bei einen der eckpunkte

> lasso essentially selects features by factoring useless features 0

Source: Saptashwa Bhattacharyya on
<https://towardsdatascience.com/ridge-and-lasso-regression-a-complete-guide-with-python-scikit-learn-e20e34bcfb0b>



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How to choose between Ridge and Lasso Regression?

- ♦ If we know that most features are usefull → Use Ridge Regression
 - ♦ If we know that many features are irrelevant → Use Lasso Regression
 - ♦ What do we do if we do not know in advance? E.g. because we have very many features?
- Combine Ridge and Lasso Regression: **Elastic Net Regression**

$$\mathbf{b} = \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \left(\frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda_1 \|\boldsymbol{\beta}\|_1 + \lambda_2 \|\boldsymbol{\beta}\|_2^2 \right)$$

how strong lasso? how strong ridge?

where λ_1 and λ_2 are positive constants.

hyperparameter search via grid search (i.e. try all combinations)

problem: constants are unlimited

Alternative form: $\mathbf{b} = \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \left(\frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \alpha \rho \|\boldsymbol{\beta}\|_1 + \alpha(1 - \rho) \|\boldsymbol{\beta}\|_2^2 \right)$ with $\alpha \geq 0$ and $0 \leq \rho \leq 1$

how much lasso? how much ridge?

puts constraints on constants

can be controlled via (tradeoff lasso/ridge) if

alpha controls how strongly model is regularized



Discussion

- ◆ Elastic Net combines the strength of Ridge and Lasso
 - $\lambda_1 = 0$ and $\lambda_2 = 0 \rightarrow$ multiple linear regression without regularization ($\alpha = 0$)
 - $\lambda_1 = 0$ and $\lambda_2 > 0 \rightarrow$ ridge regression ($\rho = 0$ and $\alpha > 0$)
 - $\lambda_1 > 0$ and $\lambda_2 = 0 \rightarrow$ lasso regression ($\rho = 1$ and $\alpha > 0$)
 - $\lambda_1 > 0$ and $\lambda_2 > 0 \rightarrow$ hybrid between lasso and ridge regression ($0 < \rho < 1$ and $\alpha > 0$)

- ◆ The hybrid works esp. good if there are correlation between the features:
 - Lasso part tends to pick one of the correlated features and eliminate the rest
 - Ridge part tends to shrink the weights of all correlated features together
 - ➔ Best of both worlds (for the right λ_1 and λ_2 / α and ρ !)



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Other Regression Algorithms

- ◆ Piecewise linear regression
- ◆ Decision Trees for Regression (“Regression Trees”)
- ◆ ANN (Artificial Neural Networks) for Regression
- ◆ SVM (Support Vector Machines)
- ◆ Gradient Boosting



Exercises

Demo and Exercise 2

Ames Housing





Key Takeaways

- ◆ Polynomial Regression
- ◆ Regularization
 - Ridge
 - Lasso
 - Elastic Net
- ◆ Further Regression Algorithms

